

The Industrialization and Economic Development of Russia through the Lens of a Neoclassical Growth Model

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Based on Cheremukhin, Golosov, Guriev, and Tsyvinski (2017)

Question and Setup

Model and data

Results

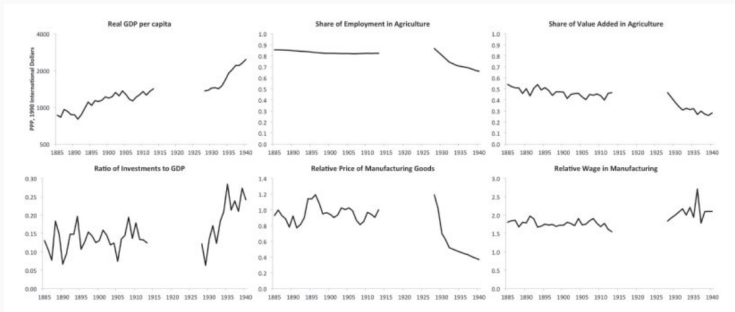
Takeaways

Question and Setup

Historical Background

- **Tsarist period (1885-1913):** economy remained predominantly agrarian, with limited labor movement to industry.
- **Institutional context:** peasant communes, high entry barriers in industry, and prevalence of cartels/monopoly power.
- **Soviet turn (1928-1940):** forced collectivization in agriculture and state-led industrial expansion under Five-Year Plans.
- **Central historical puzzle:** very rapid structural change occurred together with major productivity disruptions and severe social costs.

Figure 1: Aggregate indicators.



Model and data

- $U_{A,t}, U_{M,t}$: marginal utility of agricultural and manufacturing consumption.
- $F_{i,t}^N$: marginal product of labor .
- $F_{i,t}^K$: marginal product of capital.
- τ_{Wt} : labor wedge.
- τ_{Rt} : capital wedge.
- τ_{Kt} : intertemporal wedge.

Model Structure: Environment

Preferences:

$$\max \sum_{t=0}^{\infty} \beta^t U(c_{A,t}, c_{M,t}), \quad U = \left[\eta^{\frac{1}{\sigma}} (c_{A,t} - \gamma_A)^{\frac{\sigma-1}{\sigma}} + (1 - \eta)^{\frac{1}{\sigma}} c_{M,t}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

Production:

$$Y_{i,t} = X_{i,t} K_{i,t}^{\alpha_{K,i}} N_{i,t}^{\alpha_{N,i}}, \quad i \in \{A, M\}$$

Feasibility Constraints:

$$\underbrace{N_{A,t} + N_{M,t} = \chi_t N_t}_{\text{Labor}} \quad \text{and} \quad \underbrace{K_{A,t} + K_{M,t} = K_t}_{\text{Capital}}$$

$$N_t c_{A,t} + E_{A,t} = Y_{A,t}$$

$$N_t c_{M,t} + I_t + G_{M,t} + E_{M,t} = Y_{M,t}$$

$$K_{t+1} = I_t + (1 - \delta) K_t$$

Model Structure: Frictions as Wedges

Intersectoral Labor Wedge:

$$1 + \tau_{Wt} = \frac{U_{M,t}}{U_{A,t}} \frac{F_{M,t}^N}{F_{A,t}^N}$$

Intersectoral Capital Wedge:

$$1 + \tau_{Rt} = \frac{U_{M,t}}{U_{A,t}} \frac{F_{M,t}^K}{F_{A,t}^K}$$

Intertemporal Wedge:

$$1 + \tau_{Kt} = \beta \frac{U_{M,t+1}}{U_{M,t}} (1 + F_{M,t+1}^K - \delta)$$

The Role of Prices: Labor Wedge Decomposition

$$1 + \tau_{Wt} = \underbrace{\frac{U_{M,t}/p_{M,t}}{U_{A,t}/p_{A,t}}}_{\text{Consumption}} \times \underbrace{\frac{p_{M,t}F_{M,t}^N/w_{M,t}}{p_{A,t}F_{A,t}^N/w_{A,t}}}_{\text{Production}} \times \underbrace{\frac{w_{M,t}}{w_{A,t}}}_{\text{Mobility}}$$

Economic Frictions Identified by Prices:

- **Consumption** $\neq 1$: Rationing, segmented markets, consumer access costs.
- **Production** $\neq 1$: Markups, monopoly power, industry entry barriers.
- **Mobility** $\neq 1$: Migration restrictions (communes, passports), skill acquisition.

Model parameters

Parameter	Description	Value
$\alpha_{K,A}$	Factor shares	0.21
$\alpha_{N,A}$	of the	0.60
$\alpha_{K,M}$	production	0.34
$\alpha_{N,M}$	functions	0.66
γ^A	Subsistence level	27.5
η	Asymptotic share of agriculture	0.15
β	Discount factor	0.96
σ	Elasticity of substitution	1
ρ	Intertemporal elasticity	0
δ	Depreciation	0.05

Table 1: Parameters

Identification logic:

- Quantities and prices identify incentives and marginal products.
- Wedges are residuals from equilibrium conditions (not directly observed taxes).

Potential pitfalls:

- Soviet prices and historical data can be noisy.
- Interpretation is strongest in *changes over time* and decomposition patterns.

Results

Tsarist Russia: Main Quantitative Findings

- Intersectoral distortions are very large before 1913.
- Average labor wedge is about **14** (roughly equivalent to a 1300% ad valorem tax on labor reallocation).
- Largest component is **production**, around half of total labor wedge (about 47% in logs).
- Implied non-agricultural markups are very high (around 3–3.5).
- Mobility component is positive but comparatively smaller.
- Evidence does **not** support communes as the main barrier to industrialization.

Soviet Period (1928–1940): What Changes?

- Structural transformation accelerates strongly.
- Intersectoral labor and capital wedges decline relative to Tsarist levels.
- Decline in labor wedge is driven mainly by a large fall in the **production component**.
- Estimated non-agricultural TFP declines by about **20%** in this period.
- Agricultural TFP falls early (collectivization/famine years), then partially recovers.

Figure 2: Sectoral TFPs and wedges

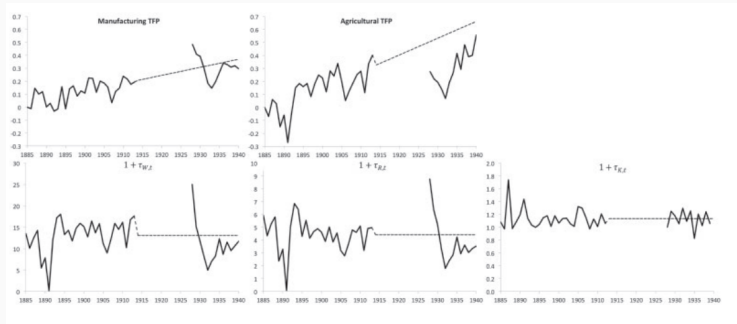
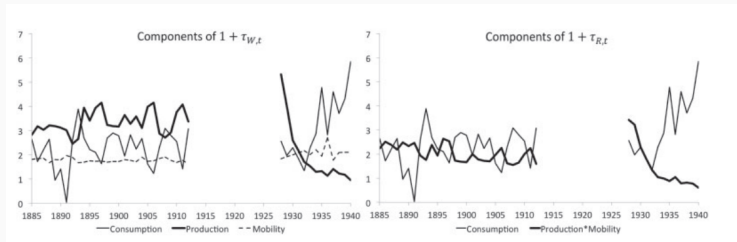


Figure 3: Wedge components



Big Push prediction (increasing returns story):

- Industrialization should raise manufacturing TFP.
- Labor wedge should increase in transition to modern equilibrium.

What the paper finds:

- Labor wedge decreases sharply.
- Manufacturing TFP does not improve; it declines over 1928–1940.

Compared with extrapolated Tsarist trends:

- Extrapolated TFP trend and fixed wedges.
- Lower labor wedge explains most extra structural change.
- Production components explain the largest part of labor reallocation.
- Sectoral TFP underperformance offsets many output gains from reduced wedges.

Counterfactual Tables

	Share of employment in agriculture	GDP per capita	Share of employment in agriculture	GDP per capita
	1939 level		1928–39 change	
	p.p.	%	p.p.	%
1. Agricultural TFP, X_A	4.9	−19.8	3.2	−6.7
2. Manufacturing TFP, X_M	−0.7	−2.6	2.0	−19.3
3. Labour distortion, τ_W	−6.2	13.2	−18.8	37.4
4. Capital distortion, τ_R	0.3	0.3	0.3	4.2
5. Investment distortion, τ_K	−1.9	3.2	−2.3	3.9
6. Defense spending	−1.3	3.7	−2.1	5.3
7. Foreign trade	−3.0	5.7	−0.6	1.1
8. Population growth	−0.5	1.5	−0.5	1.4
9. Expectations, κ	−2.1	5.6	−2.1	5.5
10. Capital accumulation, K_0	1.3	−4.7	2.4	4.8
Total	−9.2	6.1	−18.5	37.6

Counterfactual Tables

TABLE 3
Counterfactual analysis 1928–39

	Share of employment in agriculture	GDP per capita	Share of employment in agriculture	GDP per capita
	1939 level		1928-39 change	
	p.p.	%	p.p.	%
Consumption component	6.5	0.0	4.1	5.0
Production components	−15.9	20.6	−25.4	42.1
Mobility component	3.6	−7.0	2.8	−5.5

Takeaways

1. The main pre-WWI barrier to Russian industrialization was **production-side distortion**, not primarily mobility barriers from communes.
2. Soviet-era structural transformation was driven mainly by a fall in the production component of wedges.
3. Rapid industrial reallocation did not imply high productivity growth; TFP performance was weak in both sectors.
4. Prices are crucial for identifying mechanisms behind structural transformation.